

Meghana Kalikota

Python Developer

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Professional Summary:

- **Experienced Python Developer** with over **4+ years** of expertise in **software development, backend and fullstack development, and automation.**
- Strong experience in **CI/CD pipelines, API development, containerization, and cloud computing (AWS, Azure, GCP).**
- Extensive work in **microservices architecture, distributed computing, big data technologies (Hadoop, Spark, Databricks), and real-time data streaming (Kafka, RabbitMQ).**
- Expertise in **database management (SQL: MySQL, PostgreSQL | NoSQL: MongoDB, Cassandra, Redis), data modeling, and high-performance computing.**
- Strong foundation in **Linux system administration, automation (Terraform, Ansible, Bash Scripting), and IT operations.**
- Proven experience in **building AI-powered applications, optimizing LLM fine-tuning and inference, and implementing real-time RAG pipelines** for enhanced knowledge retrieval.
- Skilled in **Python, Node.js, and Java** for developing scalable backend solutions. Proficient in **real-time data ingestion, ETL pipelines, and cloud-based IoT architectures** using **AWS Lambda, Timestream, and S3.**
- Hands-on experience with **front-end development (React, Angular, Svelte, Ember.js), RESTful APIs, GraphQL, and server-side rendering (SSR, SSG).**
- Proven ability to work in **cross-functional teams, collaborate with business stakeholders, and deliver highquality, scalable software solutions.**
- Passionate about **solving complex problems, improving application performance, and driving innovation** in financial and enterprise applications.

Technical Skills:

- **Programming Languages:** Python (Django, FastAPI, Flask), Java, JavaScript (React, Angular, Svelte, Ember.js), HTML5, CSS3
- **Frameworks & Libraries:** Django, FastAPI, Flask, Pandas, NumPy, Spring Boot, TensorFlow, PyTorch
- **Cloud Platforms:** AWS (EC2, S3, Lambda, RDS, ECS), Azure (Functions, DevOps, Synapse Analytics), GCP (Cloud Functions, Vertex AI)
- **Databases:** SQL (MySQL, PostgreSQL, SQLAlchemy), NoSQL (MongoDB, Firebase, Cassandra, CosmosDB)
- **Big Data & Streaming:** Hadoop, Spark, Databricks, Kafka, RabbitMQ
- **DevOps & CI/CD:** Docker, Kubernetes, Jenkins, Bitbucket Pipelines, GitHub Actions, Terraform, Ansible
- **Monitoring & Security:** Grafana, Prometheus, CloudWatch, SSI environments, Cybersecurity best practices
- **Version Control & Collaboration:** Git, GitHub, Bitbucket, Jira, Agile/Scrum methodologies
- **Web & API Development:** RESTful APIs, GraphQL, Server-side Rendering (SSR), Static Site Generation (SSG)
- **Industrial IoT & Automation:** MQTT, OPC-UA, AWS IoT, Google Cloud IoT

Professional Experience:

Centene Corporation, Irving, TX (02/2024 - present)

Full Stack Python Developer

- Designed and developed **scalable and secure RESTful APIs** using **FastAPI, Flask, and Django REST Framework (DRF)**, improving service modularity.
- Developed, tested, and deployed **Python-based full-stack applications**, collaborating with **Agile teams** to deliver high-quality software solutions.

- Designed and implemented **scalable microservices architectures**, utilizing **Docker and Kubernetes** for seamless deployment and container orchestration.
- Managed and optimized virtual machines using **GCP Compute Engine**, configuring auto-scaling and load balancing with **GCP Cloud Load Balancing**.
- Built and maintained **CI/CD pipelines** using **Jenkins, GitHub Actions, and Terraform**, automating deployments and improving release cycles.
- Built and **integrated ML models** into production systems for real-time predictions using **AWS SageMaker** or **Google AI Platform**.
- Performed data cleaning, preprocessing, and feature extraction using **Pandas, NumPy, and scikit-learn** to ensure high-quality input for **ML models**.
- Developed **custom fine-tuned LLMs** using Hugging Face Transformers and open-source models for domain-specific applications.
- Engineered **cloud-based solutions** on **AWS**, working with services like **Lambda, API Gateway, EC2, S3, RDS, DynamoDB, and IAM** to optimize performance and scalability.
- Integrated **NoSQL (MongoDB, Cassandra)** and **SQL (PostgreSQL, MySQL)** databases to support high- performance, data-driven applications.
- Designed, developed, and maintained **scalable full-stack applications** using **TypeScript, Node.js, and JavaScript frameworks (React, Angular, NestJS)**.
- Built and optimized **RESTful and GraphQL APIs** with **TypeScript and Node.js**, ensuring high performance and scalability.
- Automated **artifact deployment to Nexus** using **Gradle/Maven**, ensuring a secure and efficient release pipeline.
- Developed and maintained **Gradle/Maven build scripts** to automate dependency management, testing, and packaging.
- Optimized **Gradle builds** using **parallel execution, caching, and dependency resolution strategies** for faster compilation.
- Implemented **data modeling and complex SQL queries** to enhance database efficiency and query performance.
- Stayed ahead of **technology trends**, actively experimenting with **Python frameworks, cloud technologies, and DevOps tools** to improve application efficiency.
- Advocated **Agile best practices**, contributing to **Scrum/Kanban teams**, leading sprint planning, and ensuring continuous delivery.
- Contributed to **technology communities**, mentoring junior engineers, sharing insights, and participating in discussions on **Python** development and cloud computing.
- Designed and implemented **ETL pipelines** to process, transform, and load large-scale datasets efficiently.
- Developed **data ingestion frameworks** using **Python, Apache Spark, and AWS Glue**, optimizing data processing workflows.
- Automated ETL workflows using **Airflow, Prefect, and Step Functions**, ensuring data integrity and monitoring.
- Created and presented detailed reports and dashboards using **Tableau, Power BI, or similar tools**, providing insights for business decision-making.
- Developed **custom data visualizations, KPIs, and drill-down reports** to provide real-time insights for decision making.

- Designed and optimized **database queries and procedures** in **PostgreSQL** to improve query performance.
- **Developed full-stack applications** using **Java (Spring Boot), Angular, and Python**, improving system efficiency.
- Developed and maintained **test automation frameworks** using **Python** and **pytest** to enhance software reliability.

Franklin University, Columbus, Ohio

Python Developer

(08/2023 - 02/2024)

- Developed a **full-stack web application** using **Python (Flask)** and **JavaScript (React/Angular)** for specific functionality.
- Designed and implemented **RESTful APIs** to enable seamless communication between the frontend and backend.
- Built an **ETL pipeline using Python and AWS Redshift** to process and analyze large datasets efficiently.
- Developed **interactive dashboards using Tableau/Power BI**, visualizing complex datasets for better insights.
- Deployed the application on **AWS (EC2, Lambda, RDS, S3, API Gateway)**, ensuring high availability and scalability.
- Automated **CI/CD pipelines using Jenkins and GitHub Actions**, reducing deployment time.
- Containerized the application using **Docker and Kubernetes**, ensuring portability and efficient resource allocation.
- Implemented **AWS IAM roles and security best practices** to restrict unauthorized access.
- Designed and optimized **SQL queries** for PostgreSQL/Redshift to improve query performance.
- Created **Kubernetes (EKS, GKE) deployment manifests and Helm charts** to manage containerized Python applications in cloud environments.
- Monitored application health and performance using **Python-based observability tools** such as **Prometheus, Grafana, ELK (Elasticsearch, Logstash, Kibana), and AWS CloudWatch**.
- Conducted **unit testing and integration testing** using **PyTest and Selenium**, improving code reliability.
- Developed **machine learning models using TensorFlow/Scikit-learn** for prediction/classification task.
- Integrated **third-party APIs** (Stripe, Twilio, OpenAI, etc.) to enhance project functionality.
- Implemented **serverless computing using AWS Lambda**, reducing infrastructure costs and improving efficiency.
- Followed **Agile methodologies (Scrum, Kanban)** and used **JIRA/Trello** for project management.
- Wrote detailed **technical documentation and user manuals**, improving project maintainability.
- Built **RAG-based AI solutions** combining LLMs with vector databases (Pinecone, FAISS, Weaviate, ChromaDB) for contextual and knowledge-aware responses.
- Developed and deployed Python applications on **AWS, Azure, and GCP**, leveraging **EC2, Lambda, API Gateway, RDS, DynamoDB, and S3** to build resilient cloud-based architectures.
- Developed **responsive and dynamic UI components** using **React.js and Angular**, improving user experience.
- Utilized **React Hooks and Redux Toolkit** for efficient state management and performance optimization.
- Implemented **lazy loading and code-splitting** in React.js/Angular to enhance application performance.
- Integrated frontend applications with **RESTful APIs** for seamless data fetching and real-time updates.
- Designed and optimized **SQL/NoSQL databases (PostgreSQL, MySQL, MongoDB, DynamoDB)** with **Python ORMs (SQLAlchemy, Django ORM, Pydantic)** to efficiently store and retrieve structured and semi-structured data.
- Automated **database migrations and schema management** using Python-based tools such as Alembic and Flyway.
- Developed **unit and integration tests** using **PyTest, Unittest, and Selenium**, improving code reliability.
- Automated **API testing with Postman, PyTest, and Newman CLI**, ensuring robust backend performance.
- Implemented **load testing and performance monitoring** using Python-based tools like Locust and JMeter.

Kony Inc, Hyderabad, India

Python Developer, SDE

(09/2020 – 12/2022)

- Developed and maintained **cloud-based applications** using **Python (Django, FastAPI, Flask)** and **JavaScript (React, Angular)**.
- Implemented **Terraform-based infrastructure provisioning**, streamlining cloud resource management.
- Monitored and logged application performance using **Datadog, ELK, and Splunk**, ensuring system reliability.
- Built **event-driven AWS Lambda functions** to process IoT sensor data in real time.

- Configured and maintained **Sonatype Nexus Repository** for hosting internal/external dependencies and artifacts.
- Integrated **Nexus with Jenkins, GitHub Actions, and AWS CodeBuild** to streamline artifact management in CI/CD.
- Developed and tested **CI/CD pipelines** within IDEs, integrating **unit testing, linting, and build automation**.
- Built **event-driven microservices** using **Spring Boot** and **Kafka** for real-time data streaming.
- Developed **FastAPI-based RESTful APIs** for fleet management and real-time vehicle tracking.
- Implemented **batch and stream processing** solutions using **Apache Spark** and **PySpark** for scalable
- Developed **server-side applications** using **C# (.NET Core, ASP.NET)** for enterprise solutions, integrating with databases and cloud services.
- Conducted **product testing, usability testing, and UAT** using **Selenium and automation frameworks**, ensuring quality releases.
- Integrated **Dynatrace with Kubernetes (EKS, GKE) and Docker**, providing real-time insights into containerized workloads.
- Developed and integrated **AI/ML models** and **LLM-based solutions** to enhance automation, data analysis, and intelligent decision-making in applications.
- Built and optimized **REST APIs**, integrating them with **frontend components** for seamless user experiences.
- Designed and implemented **big data solutions** using **Apache Spark, Hadoop, and Azure Data Services**.
- Built and optimized **high-performance backend microservices** in **Python and C++**, leveraging **multithreading and asynchronous processing** for large-scale data applications.
- Designed and implemented **RESTful APIs and GraphQL endpoints** for seamless integration between frontend applications and cloud-based backends.
- Developed and optimized complex SQL queries for **data extraction, transformation, and reporting**, ensuring data accuracy and integrity across various **RDBMS** such as **MySQL, PostgreSQL, SQL Server, and Oracle**.
- Designed and implemented **database schemas** based on **data modeling principles**, ensuring adherence to **normalization standards** for efficient data storage.
- Led the automation of data testing processes using **automation frameworks** and tools, significantly reducing manual testing efforts and increasing overall testing efficiency.
- Conducted **performance tuning** and **query optimization** to enhance database efficiency, which resulted in reduced processing time for large datasets.
- Collaborated with cross-functional teams to perform **data quality assessments**, identifying discrepancies and implementing corrective actions.
- Mentored junior team members in best practices for **SQL development, data modeling, and performance tuning**, fostering a collaborative and knowledge-sharing environment.
- Developed comprehensive **documentation** for data processes, including ETL workflows, test cases, and data lineage for compliance and audit purposes.
- Gained exposure to **Terraform and infrastructure automation** to improve deployment efficiency.
- Followed **Agile methodologies**, participating in **sprint planning, daily stand-ups, and retrospectives**.
- Ensured **code quality, performance, and scalability** by applying **design patterns and best practices**.
- Provided application **development and production support**, troubleshooting and resolving system issues.
- Worked in a **global delivery environment**, coordinating with **offshore and onsite teams**
- Designed and implemented **custom XML schema definitions (XSD)** for structured data processing in enterprise applications.
- Integrated **PHP-based web applications** with MySQL and PostgreSQL, ensuring efficient database operations.
- Used **Axios and Fetch API** for efficient API calls and handled request/response lifecycle effectively.
- Designed **server-side rendered (SSR) applications** with **SvelteKit**, reducing **bundle size** and improving **SEO performance**.

ECIL, Hyderabad, India

(01/2020 – 06/2020)

Python Intern

- Assisted in **Python application development** and debugging.
- Worked with **Pandas and NumPy** for data processing and analysis.
- Built basic **web scraping applications** using **BeautifulSoup and Selenium**.
- Gained experience with **Git version control**, collaborating on projects.
- Contributed to **unit testing** using **PyTest and Unittest**.

Education:

Franklin University – Master's in Information Technology (Cyber Security)

Jawaharlal Nehru Technological University – Bachelor's in Computer Science